TECHMINDS ACADEMY

JavaScript Fundamentals

Web Interactivity Curriculum · Beginner to Job-Ready

A structured 14-week programme for students who have completed the TechMinds HTML and CSS curricula. From your first line of JavaScript to building a fully interactive, API-powered web application.

14	5	65+	9	2
Weeks	Stages	Lessons	Projects	Prerequisites

DOM Manipulation · Control Flow · Arrays & Objects · Async/Await · Fetch API · ES6+ Modules · Capstone App

LEARNING PATHWAY

Programme Overview

Stage	Focus	Duration	Level	Prerequisite
Stage 1	JS Foundations — variables, DOM, events	3 weeks	Beginner	HTML + CSS
Stage 2	Control Flow & Functions	3 weeks	Beginner	Stage 1
Stage 3	Arrays & Objects	2 weeks	Intermediate	Stage 2
Stage 4	Async JavaScript & APIs	3 weeks	Intermediate	Stage 3
Stage 5	ES6+ Modules & Capstone App	3 weeks	Advanced	Stage 4

CURRICULUM RATIONALE

Key Decisions Explained

Why JavaScript third?	JavaScript adds behaviour to structure (HTML) and style (CSS). A student who jumps into JS without HTML and CSS foundations will struggle to see results. The TechMinds sequence ensures every concept builds naturally on the last.
Why teach the DOM first?	Beginners need to see results immediately. Teaching DOM manipulation in Week 3 means the student is already making real web pages interactive before they reach functions or arrays. Motivation stays high throughout.
Why dedicate 3 weeks to async?	Async JavaScript is the concept that breaks most beginners. Three weeks allows the student to fully understand callbacks, promises, and async/await — not just copy code they don't understand.
What comes after this course?	JavaScript Fundamentals feeds into the TechMinds React.js course — the next step in the web development track. Students who complete this programme are fully prepared to enter the React ecosystem.

Stage 01 · Beginner 3 Weeks · Weeks 1 – 3

JavaScript Foundations

Variables, data types, and your first interactive web page

This stage assumes the student has completed the HTML and CSS curricula. JavaScript is introduced from scratch — what it is, how the browser runs it, and how to write code that responds to user actions. By the end of Week 3 the student will be making their web pages interactive.

Week 1 — Introduction to JavaScript

5 lessons · 6 hrs

01 What is JavaScript? What does it do in the browser?

02 Adding JS to a page: inline, internal, and external scripts

03 The browser console — your development tool

04 Variables: var, let, and const — differences and best practices

05 Data types: string, number, boolean, null, undefined

06 console.log() — printing values and debugging

PROJECT A console program that stores your name, age, and city and prints a greeting

Week 2 — Operators & Expressions

5 lessons · 6 hrs

01 Arithmetic operators: +, -, *, /, %, **

02 Assignment operators: =, +=, -=, *=, /=

03 Comparison operators: ==, ===, !=, !==, >, <, >=, <=

04 Logical operators: && (AND), || (OR), ! (NOT)

05 The ternary operator — shorthand if/else

06 Template literals — embedding variables in strings

PROJECT A tip calculator — input a bill amount and percentage, output the tip and total

Week 3 — DOM Manipulation — Making Pages Interactive

5 lessons · 7 hrs

01 What is the DOM? How JavaScript sees the HTML page

02 Selecting elements: getElementById, querySelector

03 Reading and changing text with textContent and innerHTML

04 Changing CSS styles with element.style

05 Adding and removing CSS classes with classList

06 Event listeners — responding to clicks and keypresses

PROJECT A colour switcher — click buttons to change the page background colour

Stage 02 · Beginner 3 Weeks · Weeks 4 – 6

Control Flow & Functions

Making decisions, repeating actions, and writing reusable code

Control flow and functions are the building blocks of every program. This stage teaches the student to write code that makes decisions, repeats actions, and reuses logic without repetition. Every concept is applied immediately to real DOM interactions.

Week 4 Conditionals

5 lessons · 6 hrs

01 if, else if, else — writing conditional logic

02 Truthy and falsy values — what JavaScript considers true

03 The switch statement — cleaner multi-branch logic

04 Nested conditionals — conditions inside conditions

05 Short-circuit evaluation with && and ||

06 Practical DOM example: show and hide elements conditionally

PROJECT A quiz app — 3 questions, check answers, show a score at the end

Week 5 Loops

5 lessons · 6 hrs

01 The for loop — counting and iterating

02 The while loop — repeat until a condition is false

03 The do...while loop

04 break and continue — controlling loop execution

05 Looping through arrays with for...of

06 Building a list dynamically in the DOM using a loop

PROJECT A number guessing game — randomly pick a number, loop until the user guesses correctly

Week 6 Functions

5 lessons · 7 hrs

01 Declaring and calling functions

02 Parameters and arguments — passing data into functions

03 Return values — getting data back from functions

04 Arrow functions: () => {} syntax

05 Scope: local vs global variables

06 Higher-order functions — passing functions as arguments

PROJECT A unit converter — functions for km to miles, kg to lbs, °C to °F, triggered by buttons

Stage 03 · Intermediate 2 Weeks · Weeks 7 – 8

Arrays & Objects

Working with collections and structured data

Real applications store and process data. This stage introduces arrays and objects — the two fundamental data structures in JavaScript. Students learn to create, read, update, and delete data, and to render that data dynamically into the DOM.

Week 7 **Arrays** 5 lessons · 6 hrs

01 Creating arrays and accessing items by index
02 Array methods: push, pop, shift, unshift, splice
03 Iterating arrays: forEach, map, filter, find, reduce
04 Spread operator and array destructuring
05 Sorting and searching arrays
06 Rendering an array as a list of DOM elements

PROJECT A to-do list — add, display, and remove tasks stored in an array

Week 8 **Objects** 5 lessons · 6 hrs

01 Creating objects with key-value pairs
02 Accessing and updating object properties: dot and bracket notation
03 Object methods — functions inside objects
04 Object destructuring
05 Arrays of objects — the real-world data pattern
06 Rendering an array of objects as cards in the DOM

PROJECT A product listing page — an array of product objects rendered as styled cards

Stage 04 · Intermediate 3 Weeks · Weeks 9 – 11

Async JavaScript & APIs

Fetching live data and handling asynchronous operations

Every modern web application fetches data from a server. This stage demystifies asynchronous JavaScript — promises, async/await, and the fetch API. Students will connect their pages to real public APIs and display live data, which is one of the most sought-after junior skills.

Week 9 — Asynchronous JavaScript & Promises

5 lessons · 7 hrs

- 01 Synchronous vs asynchronous code — why it matters
- 02 setTimeout and setInterval — scheduling code
- 03 Callbacks — the original async pattern
- 04 Promises — a cleaner way to handle future values
- 05 .then() and .catch() — chaining promise handlers
- 06 Promise.all() — running multiple async operations at once

PROJECT A countdown timer — counts down from 10 and displays a message when done

Week 10 — async/await & the Fetch API

5 lessons · 7 hrs

- 01 async/await — writing asynchronous code that reads like synchronous
- 02 try/catch — handling errors in async functions
- 03 The fetch() function — making HTTP GET requests
- 04 Parsing JSON responses with response.json()
- 05 Displaying fetched data in the DOM
- 06 Loading states — showing a spinner while data loads

PROJECT A weather app — fetch live weather from a public API and display temperature and conditions

Week 11 — Working with Real APIs

5 lessons · 7 hrs

- 01 Reading API documentation — endpoints, parameters, keys
- 02 Query parameters — filtering and searching API results
- 03 Error handling — what to show when the API fails
- 04 Debouncing — preventing too many API calls on keypress
- 05 Local storage — caching API results in the browser
- 06 POST requests — sending data to an API with fetch

PROJECT A movie search app — search the OMDB API, display posters, titles, and ratings

Stage 05 · Advanced 3 Weeks · Weeks 10 – 12

ES6+, Modules & Capstone

Modern JavaScript and building a complete interactive web application

The final stage covers the modern JavaScript features every developer must know — ES6+ syntax, modules, and clean code patterns. The course closes with a capstone project: a fully interactive, API-powered web application built with HTML, CSS, and JavaScript.

Week 10 — Modern JavaScript — ES6+ Features

5 lessons · 6 hrs

01 let and const — block scoping revisited
02 Destructuring: arrays and objects in depth
03 The spread and rest operators
04 Default function parameters
05 Optional chaining (?.) and nullish coalescing (??)
06 Array methods: flat, flatMap, Object.keys, Object.entries

PROJECT Refactor a previous project to use modern ES6+ syntax throughout

Week 11 — JavaScript Modules & Code Organisation

5 lessons · 6 hrs

01 What are modules and why do they matter?
02 export and import — splitting code across files
03 Default exports vs named exports
04 Organising a project: separating data, logic, and UI
05 Introduction to npm and using third-party packages
06 Intro to build tools: what Vite does and why it exists

PROJECT Split the movie search app into modules: api.js, render.js, and main.js

Week 12 — Capstone Project

6 sessions · 12 hrs

01 Planning the app — features, API choice, wireframe
02 Project setup — file structure, modules, and stylesheet
03 Building the UI — HTML structure and CSS layout
04 Connecting the API — fetch, parse, and render data
05 Adding interactivity — search, filter, and local storage
06 Final polish — error handling, loading states, and deployment

PROJECT A complete interactive web app — student chooses: recipe finder, GitHub profile viewer, or crypto tracker

ASSESSMENT & COMPLETION

How Students Are Evaluated

Component	Description	Weight
Weekly Projects (×9)	Interactive feature or app built from scratch each week	**40%**
Stage Assessments (×5)	Short quiz and code review at end of each stage	**30%**
Capstone Application	Full interactive API-powered app — built, deployed, and reviewed	**30%**

LEARNING OUTCOMES

What Students Can Do on Completion

On completing this programme, the student will be able to:

- Write clean, modern JavaScript (ES6+) from scratch
- Select and manipulate DOM elements to build interactive web pages
- Use conditionals, loops, and functions to control program logic
- Store and process data using arrays and objects
- Write asynchronous code confidently with async/await and try/catch
- Fetch and display live data from real public REST APIs
- Organise code into ES6 modules with import and export
- Build and deploy a complete interactive web application
- Read and understand third-party API documentation independently

TOOLS & RESOURCES

Everything the Student Needs

Tool	Purpose	Cost
VS Code	Primary code editor for writing JavaScript	Free
Live Server	Auto-refreshes browser on every file save	Free
Chrome DevTools	Console, debugger, network tab for API calls	Built-in
Node.js + npm	Run JavaScript outside the browser; install packages	Free
Vite	Fast build tool for working with ES6 modules	Free
Public APIs	OpenWeather, OMDB, REST Countries — used in projects	Free tier

GitHub Pages	Deploy and share the capstone app publicly	Free